NETFLIX DATA ANALYTICS PROJECT

A Study Using Excel, SQL, Python and Power BI

COLLEGE PROJECT REPORT
Submitted in partial fulfillment of the requirements for the
Bachelor of Computer Applications (BCA)

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Department: BCA
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CERTIFICATE

This is to certify that Alok Kumar Singh, a student of Bachelor of Computer Applications (BCA) at Jeevandeep Mahavidyalaya, has completed the project titled “Netflix Data Analytics Using Excel, SQL, Python and Power BI” under the guidance of Bhupendra bharadwaj.

This project is submitted as part of the academic requirements of the BCA programme.
Project Guide: Bhupendra bharadwaj Head of Department: Vikram Pratap Singh Date:
Note: Obtain the required approval and signature from the college before submission.

DECLARATION

I hereby declare that the project titled “Netflix Data Analytics Using Excel, SQL, Python and Power BI” has been prepared by me for academic purposes. The sources used in this project have been acknowledged in the references section.

Student Name: Shreya Singh

Signature: _____

Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my teachers and project guide for their guidance and support during this project. I am thankful to my college for providing me with the opportunity to work on this data analytics project. This project helped me improve my practical knowledge of Excel, SQL, Python and Power BI. I also acknowledge the dataset and documentation resources used during the project.

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ABSTRACT

This project analyses a Netflix titles dataset using Microsoft Excel, SQL, Python and Power BI. The workflow includes inspecting the raw data, preparing a cleaned dataset, querying the data, creating additional analytical fields and presenting information through a dashboard.

The repository contains an Excel workbook, an uncleaned CSV, a cleaned CSV, a Python script, an SQL script and a Power BI PBIX file. The Python script uses Pandas and NumPy to inspect data quality, standardize selected text columns, handle selected missing values, convert date_added to a date type and create derived fields. The SQL script includes queries for record counts, content-type counts, rating-related filtering and country-wise summaries.

The report documents the project workflow and explains the purpose of the tools. Dataset totals and final chart observations should be confirmed from the local files and dashboard before submission; this report does not invent numerical findings that were not verifiable from the repository preview.

CHAPTER 1: INTRODUCTION

Data analytics is the process of examining data to identify patterns, summarize information and communicate findings. It commonly includes data collection, cleaning, transformation, analysis and visualization.

The Netflix Data Analytics project studies a catalogue of titles with fields such as title, type, director, cast, country, date_added, release_year, rating, duration, listed_in and description. The project uses four tools to demonstrate a practical analytics workflow.

Excel supports spreadsheet inspection and exploration. SQL supports structured querying and aggregation. Python supports repeatable data preparation and feature engineering. Power BI supports visual presentation of selected measures and patterns.

1.1 Problem Statement: A dataset with many rows and columns can be difficult to understand without a structured process. Missing values, inconsistent text and different duration units can also affect analysis. This project organizes the data and uses several analytics tools to explore the available catalogue information.

1.2 Objectives: (1) inspect the original dataset; (2) prepare a cleaned copy; (3) analyse movies and TV shows; (4) use SQL to count and group records; (5) use Python for data-quality checks and feature engineering; (6) present selected measures through Power BI; and (7) document the workflow and limitations.

1.3 Scope: The project describes the records present in the selected dataset. It does not directly measure real-time Netflix availability, actual viewing behaviour, audience preferences, subscription activity or streaming revenue.

CHAPTER 2: DATASET DESCRIPTION

The repository includes an uncleaned CSV named “netflix_titles_uncleaned data.csv” and a cleaned CSV named “netflix_dataset_cleaned data.csv”. It also includes an Excel workbook named “netflix_dataset_LOAD THE DATASET.xlsx”.

The common Netflix titles dataset structure includes fields such as show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in and description. Confirm the exact column names and row count in the submitted copy of the dataset.

Table 2.1 — Dataset summary: Dataset name: Netflix Titles Dataset; raw file: netflix_titles_uncleaned data.csv; cleaned file: netflix_dataset_cleaned data.csv; spreadsheet: netflix_dataset_LOAD THE DATASET.xlsx; source repository: https://github.com/Shreyapy2005/Netflix-.College-Project

Table 2.2 — Key columns: show_id identifies a record; type identifies Movie or TV Show; title stores the title name; country stores country information; date_added stores the date added when available; release_year stores the release year; rating stores the content rating; duration stores a movie duration or number of seasons; listed_in stores category/genre text; description contains a short description.

CHAPTER 3: TOOLS AND METHODOLOGY

3.1 Microsoft Excel: The repository includes an Excel workbook. Use it to inspect rows and columns, filter records, check data quality, prepare summaries and create tables or charts where applicable.

3.2 SQL: The SQL script creates/uses a database named netflix_db and queries a table named netflix_data. The queries demonstrate SELECT, COUNT, WHERE, GROUP BY, ORDER BY, IS NOT NULL and a subquery.

3.3 Python: The Python script imports Pandas and NumPy, loads an Excel file, inspects data dimensions and types, checks missing values and duplicates, replaces selected dirty-value strings, trims whitespace, standardizes selected text fields, converts date_added and creates derived columns.

3.4 Power BI: The repository includes a PBIX dashboard file named “netflix dashboard_power BI.pbix”. Open the file in Power BI Desktop and add a screenshot of the actual dashboard to this report.
3.5 Workflow: Raw data → inspection → cleaning and preparation → Excel exploration → SQL queries → Python analysis and feature engineering → Power BI dashboard → findings and report.

CHAPTER 4: DATA CLEANING AND PREPARATION

The Python script checks the shape, column names, data types and statistical summary of the input data. It checks missing values using isnull().sum() and checks duplicate rows using duplicated().sum().

The script replaces selected text values such as N/A, NULL, empty strings and a single space with missing values. It strips leading and trailing whitespace from object-type columns and standardizes capitalization in selected text columns.

The date_added column is converted with pandas.to_datetime using errors='coerce'. Selected missing values in director, cast, country, rating and duration are filled with the text “Unknown”. This makes missingness explicit, but those values should not be treated as real categories in every analysis.

Feature engineering in the script creates Added_Year, Added_Month and Added_Quarter from date_added; Content_Year from the current year minus release_year; Duration_Value by extracting digits from duration; Duration_Unit by extracting letters from duration; Content_Category by mapping Movie to Film and TV Show to TV Series; and Release_Decade from release_year.

Important validation note: the script shown in the repository ends after preparing and printing selected columns. Confirm that the cleaned CSV was exported after all intended transformations. Also validate Duration_Unit extraction and the treatment of null dates before using these fields in charts.

Table 4.1 — Cleaning evidence to record: total rows before [enter value]; total rows after [enter value]; duplicate rows found [enter value]; missing values handled [describe actual fields]; date conversion issues [enter value]; final cleaned file [confirm filename].

CHAPTER 5: EXCEL ANALYSIS

Excel is useful for quick inspection, filtering, summary calculations and visual exploration. The repository contains a workbook named “netflix_dataset_LOAD THE DATASET.xlsx”.

Suggested report documentation for the Excel stage: describe how the data was loaded; note any cleaning checks completed in Excel; show relevant formulas, filters, Pivot Tables or charts that are actually present in the workbook; and explain what each output tells the reader.

Possible analytical questions include: How many records are present? How are records divided between Movie and TV Show? How do titles vary by release year? What content ratings appear most often? Which country values and categories occur frequently?

Do not state that a Pivot Table, formula or chart was created unless it exists in the workbook. Add Figure 5.1: Excel dataset view; Figure 5.2: summary table or Pivot Table; Figure 5.3: Excel chart, if available.

CHAPTER 6: SQL ANALYSIS

The SQL script creates a database named netflix_db, selects that database and queries a table named netflix_data. Confirm that the table name and column names match the imported data.

Query 1 — Preview records: SELECT * FROM netflix_data LIMIT 10; This displays a small sample of records.

Query 2 — Count all records: SELECT COUNT(*) AS show_id FROM netflix_data; The alias could be renamed to total_titles for clarity because the query counts rows rather than show_id values specifically.

Query 3 — Count by content type: SELECT type, COUNT(*) AS total_count FROM netflix_data GROUP BY type; This groups the records by Movie and TV Show.

Query 4 — Rating filter: the current script compares rating to MAX(rating) using a subquery. Since rating is normally a categorical text field, MAX(rating) returns the maximum value according to the database's ordering rules, not necessarily the highest audience or quality rating. Describe this query as a maximum text-value filter or replace it with an analysis appropriate to the meaning of rating.

Query 5 — Country and content type: SELECT country, type, COUNT(*) AS total_titles FROM netflix_data WHERE country IS NOT NULL GROUP BY country, type ORDER BY country, total_titles DESC; This counts records by the stored country string and type. Multiple countries stored in one field are treated as one combined value unless split into separate rows.

Query 6 — Top 10 country values among movies: SELECT country, COUNT(*) AS total_movies FROM netflix_data WHERE type = 'Movie' AND country IS NOT NULL GROUP BY country ORDER BY total_movies DESC LIMIT 10; This ranks the stored country values by movie record count. It does not necessarily count each individual country separately when a record contains multiple countries.

Insert screenshots showing the query and its result grid. Record the actual output values and briefly explain them.

CHAPTER 7: PYTHON ANALYSIS

The Python script uses Pandas for tabular data operations and NumPy for missing-value handling. It loads an Excel workbook with pd.read_excel and checks the first and last rows, shape, columns, data types and descriptive statistics.

Data quality checks include df.isnull().sum() and df.duplicated().sum(). The script replaces selected placeholder strings, strips whitespace from text columns, standardizes capitalization in several columns and converts date_added to datetime.

Feature engineering adds date-based columns, a content-age measure, numeric duration, duration unit, content category and release decade. These columns can support grouping and visualization when their definitions and missing values are checked.

Example code excerpt from the project: df['date_added'] = pd.to_datetime(df['date_added'], errors='coerce'). This converts parseable values to dates and turns invalid values into missing values.
Example code excerpt: df['Duration_Value'] = pd.to_numeric(df['duration'].str.extract(r'(\d+)')[0], errors='coerce'). This extracts the first numeric component from the duration string and converts it to a numeric value where possible.

Validation note: Content_Year uses the current year at execution time, so its values will change when the script is rerun in a later year. Duration_Value mixes minutes and seasons unless Duration_Unit is used to separate them. Verify the actual output and correct any unit-extraction issues before interpreting duration.

CHAPTER 8: POWER BI DASHBOARD

The repository includes a Power BI Desktop file named “netflix dashboard_power BI.pbix”. The dashboard should be opened in Power BI Desktop to verify its visuals and measures.

In this chapter, explain the purpose of each actual visual in the dashboard. Possible measures and visuals include total titles, movies versus TV shows, release-year distribution, country distribution, content-rating distribution and category distribution, but list only those present in the submitted PBIX file.

Describe any slicers or filters that are actually available. Explain how a user can select a category or year and observe changes in the dashboard. Verify that totals match the cleaned dataset and that null or combined country values are interpreted correctly.

CHAPTER 9: FINDINGS AND DISCUSSION

Final findings must be written after checking the outputs in Excel, SQL, Python and Power BI. Use the following structure and replace each bracket with a verified observation.

Finding 1 — Content type: The dataset contains [count] Movie records and [count] TV Show records. This indicates [describe the actual comparison].

Finding 2 — Release year: The release-year analysis shows [describe the actual pattern].

Finding 3 — Content ratings: The most frequent rating category in the analysed data is [category], with [count] records, based on [method].

Finding 4 — Country information: The most frequent stored country value is [value]. Note whether combined country strings were split or kept as recorded.

Finding 5 — Data quality: The cleaning stage identified [number] duplicate rows and [describe the important missing-value issues].

Finding 6 — Dashboard: The Power BI dashboard communicates [list the actual metrics and visuals].

Excel supports quick exploration, SQL provides structured queries, Python supports repeatable transformations and Power BI presents results visually. Using all four tools creates a clear end-to-end workflow.

CHAPTER 10: LIMITATIONS

The analysis depends on the selected dataset and its quality. Some fields may contain missing values, and multiple country or category values may be stored in one cell.

The dataset describes catalogue records; it does not directly provide real-time availability, actual viewing hours, subscriber behaviour or revenue.

Text-based ratings should be interpreted as categories unless the dataset provides a meaningful numeric rating scale. The SQL MAX(rating) query should not be described as finding the best-rated title without an appropriate numeric measure.

Derived fields such as Content_Year and Duration_Value require careful interpretation. Movie durations are generally expressed in minutes, whereas TV Show durations may represent seasons.

CHAPTER 11: CONCLUSION

This project demonstrates a practical workflow for analysing a Netflix titles dataset with Excel, SQL, Python and Power BI. The project files show work on data inspection, cleaning, SQL aggregation and filtering, Python-based feature engineering and dashboard development.

The use of multiple tools helps separate data preparation, querying, analysis and presentation. Before final submission, the numerical findings and screenshots should be added from the actual outputs, and the final dataset should be checked for consistency across tools.

The project provides practical experience in handling structured data, writing queries, preparing derived fields and communicating results through visualizations.

CHAPTER 12: FUTURE SCOPE

Future improvements may include splitting multi-country and multi-genre fields into normalized tables, creating more robust validation checks, documenting data-cleaning decisions, automating repeatable refresh steps and adding clearer dashboard filters.

If a reliable additional dataset is available, the project could be extended to study external ratings or other measures. Any new analysis should clearly identify the source and avoid assuming that catalogue data represents actual viewer behaviour.

REFERENCES

1. Project repository: https://github.com/Shreyapy2005/Netflix-.College-Project

2. Pandas documentation: https://pandas.pydata.org/docs/

3. NumPy documentation: https://numpy.org/doc/

4. MySQL documentation: https://dev.mysql.com/doc/

5. Microsoft Excel support: https://support.microsoft.com/excel

6. Microsoft Power BI documentation: https://learn.microsoft.com/power-bi/
Include the original dataset source URL here if it differs from the project repository.

APPENDIX

Appendix A — Full SQL script: refer to the file “netflix project sql..sql” in the project repository.

Appendix B — Python script: refer to “netflix_project_python.py”.

Appendix C — Dataset files: refer to “netflix_titles_uncleaned data.csv” and “netflix_dataset_cleaned data.csv”.

Appendix D — Power BI dashboard: refer to “netflix dashboard_power BI.pbix”.

Appendix E — Add screenshots of the Excel workbook, SQL outputs, Python outputs and Power BI dashboard before submitting the final report.